PROBLEM LOOKUP — READ FIRST

Question asks for...	Go to
power density $S(R,\theta)$ from a dipole	DIPOLE TYPE $\rightarrow$ HERTZIAN or ARB
HPBW β from $F(\theta)$	BEAM PARAM step 2
solid angle Ω_p / directivity D	BEAM PARAM step 3–4
$\theta_{\max}$, $S_{\max}$, plot $F(\theta)$	ARB-LENGTH DIPOLE
effective area A_e / ratio A_e/A_p	EFFECTIVE AREA
P_r given P_t, G_t, G_r, R (link)	FRIIS FORMULA

Step 0 always: compute $\lambda = c/f$. Step 1: classify dipole by l/λ.

DIPOLE TYPE BY LENGTH

Compute first *$\lambda = c/f$, then ratio l/λ*

l/λ	Type	Use formula
$< 1/50$	Hertzian (short)	HERTZIAN $S(R,\theta)$, $D=1.5$
$= 1/4$	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
$= 1/2$	Half-wave	half-wave shortcut, $D=1.64$, $R_{\rm rad}=73\,\Omega$
$= 1$	Full-wave	ARB, $D = 2.41$, $R_{\rm rad}=199\,\Omega$
other	Arbitrary	ARB formula

If $l/\lambda < 1/50$ STOP and use Hertzian. Do NOT plug small l into the arbitrary formula — you'll get the wrong shape.

HERTZIAN DIPOLE — $S(R,\theta)$

Power density *$l/\lambda < 1/50$; far field*

$$S(R,\theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2}\sin^2\theta \ (\mathrm{W/m^2})$$

$k = 2\pi/\lambda$ (rad/m); I_0 peak current (A); l length (m); R obs. dist (m).

Max direction *broadside* $\theta = \pi/2$

Total radiated power *integrate over 4π*

$$P_{\rm rad} = \frac{\eta_0\pi}{3}\left(\frac{I_0 l}{\lambda}\right)^2 \ (\mathrm{W})$$

Directivity *exact* $D_{\rm Hertz} = 1.5$ (1.76 dB)

Example (HW4 P1): l = 1m, f = 1MHz, I_0 = 12A, R = 5km, $\theta = 45°$. λ = 300m, $l/\lambda = 1/300 < 1/50$ ✓.

$S = \frac{(120\pi)(2\pi/300)^2(144)(1)}{32\pi^2(5\times10^3)^2}\cdot\frac{1}{2} = \mathbf{1.51\times10^{-9}}$ W/m^2.

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density *any l, at distance R*

$$S(\theta) = \frac{15 I_0^2}{\pi R^2}\left[\frac{\cos\left(\frac{\pi l}{\lambda}\cos\theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin\theta}\right]^2$$

Normalized pattern *dimensionless* $F(\theta) = S(\theta)/S_{\max}$

Half-wave $l = \lambda/2$ *$\pi l/\lambda = \pi/2$, shortcut form*

$$S(\theta) = \frac{15 I_0^2}{\pi R^2}\,\frac{\cos^2\left(\frac{\pi}{2}\cos\theta\right)}{\sin^2\theta}$$

$D = 1.64$, $R_{\rm rad} = 73\,\Omega$, $\theta_{\max} = \pi/2$.

Quarter-wave $l = \lambda/4$ *$\pi l/\lambda = \pi/4$, $\cos(\pi/4) = \sqrt{2}/2$*

$$\frac{S(\theta)}{S_0} = \left[\frac{\cos\left(\frac{\pi}{4}\cos\theta\right) - \frac{\sqrt{2}}{2}}{\sin\theta}\right]^2, \ S_0 = \frac{15 I_0^2}{\pi R^2}$$

$$S_{\max} = S(\pi/2) = S_0\left(1 - \frac{\sqrt{2}}{2}\right)^2 \approx 0.0858\,S_0$$

HW4 P3: $\theta_{\max} = \pi/2$ (broadside, same as half-wave); $S_{\max} = 0.0858\,S_0$.

BEAM PARAMETERS — HPBW, Ω_p, D

Step 1 — Normalize *always start here*

$$F(\theta) = S(\theta)/S_{\max}$$

Step 2 — HPBW β *solve $F(\theta_{1/2}) = 0.5$, then $\beta = 2\theta_{1/2}$ (symmetric beams)*

Gaussian shortcut: $F = e^{-a\theta^2} = 0.5 \Rightarrow \theta_{1/2} = \sqrt{\ln 2/a}$.

HW4 P2: $F = e^{-20\theta^2}$, $a = 20 \Rightarrow \theta_{1/2} = \sqrt{\ln 2/20} = 0.186$ rad, $\beta = 0.372$ rad $= 21.31°$.

Step 3 — Solid angle Ω_p *(sr)*

$$\Omega_p = \int_0^{2\pi}\int_0^{\pi} F(\theta)\sin\theta\,d\theta\,d\phi$$

If F has no ϕ dep.: $\Omega_p = 2\pi\int_0^\pi F(\theta)\sin\theta\,d\theta$.

Most integrals have NO closed form. Use Wolfram / numerical integration.

HW4 P2: $\Omega_p = 2\pi\int_0^\pi e^{-20\theta^2}\sin\theta\,d\theta = \mathbf{0.156}$ sr (numeric).

Step 4 — Directivity

$$D = \frac{4\pi}{\Omega_p} \ \text{(unitless)} \qquad D_{\rm dB} = 10\log_{10} D$$

HW4 P2: $D = 4\pi/0.156 = \mathbf{80.55}$ (19.06 dB), very directive.

Step 5 — Gain *ξ = radiation efficiency, $0 < \xi \le 1$*

$$G = \xi D \qquad G_{\rm dB} = 10\log_{10} G$$

Lossless / ideal antenna $\Rightarrow G = D$.

COMMON DIPOLE PARAMETERS

Type	D	$D_{\rm dB}$	$R_{\rm rad}$ (Ω)
Isotropic	1	0	—
Hertzian	1.5	1.76	$80\pi^2(l/\lambda)^2$
Half-wave	1.64	2.15	73
$l=\lambda$	2.41	3.82	199

Half-wave is the default "dipole antenna" unless stated otherwise.

EFFECTIVE AREA A_e

Definition *antenna's RX cross section (m^2)*

$$A_e = \frac{\lambda^2 D}{4\pi} \ (\mathrm{m^2})$$

Physical cross section *wire dipole, diameter d*

$$A_p = l\,d = \frac{\lambda}{2}\,d \ \text{(half-wave)}$$

Inverse: D from A_e

$$D = \frac{4\pi A_e}{\lambda^2}$$

HW4 P4: half-wave, $f = 100$MHz, $d = 2$cm. $\lambda = 3$m, $D = 1.64 \Rightarrow A_e = \frac{9(1.64)}{4\pi} = \mathbf{1.17}$ m^2. $A_p = (1.5)(0.02) = 0.03$ m^2. Ratio $\mathbf{39.2}\times$.

Thin wires: $A_e \gg A_p$. Antenna captures field over $\sim\lambda$ region.

POWER FLOW IDENTITIES

$$P_{\rm rad} = S_{\max}\,\Omega_p\,R^2 = \frac{4\pi R^2 S_{\max}}{D}$$

$$S_{\max} = \frac{P_{\rm rad}\,D}{4\pi R^2} \ (\mathrm{W/m^2})$$

Isotropic radiator: $S = P_{rad}/(4\pi R^2)$ (no D factor).

FRIIS TRANSMISSION FORMULA

Far-field link budget *P_t tx, P_r rx, range R*

$$\boxed{P_r = G_t G_r\left(\frac{\lambda}{4\pi R}\right)^2 P_t} \ (\mathrm{W})$$

G_t, G_r linear gains; $\lambda = c/f$; R in m; P_t, P_r in W.

Solve for R *max range*

$$R = \frac{\lambda}{4\pi}\sqrt{\frac{G_t G_r P_t}{P_r}} \ (\mathrm{m})$$

Solve for P_r **using** A_e *equivalent form*

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, $P_r = G_t G_r P_t / L_{fs}$.

Use linear G. Convert dB first: $G_{lin} = 10^{G_{dB}/10}$.

HW4 P5: half-wave TX, P_t = 1kW, f = 50MHz, RX G_r = 3dB = 2, R = 30km. λ = 6m, G_t = 1.64. $P_r = (1.64)(2)\left(\frac{6}{4\pi 3\times10^4}\right)^2 10^3 = \mathbf{8.3\times10^{-7}}$ W (0.83 μW).

dB $\leftrightarrow$ LINEAR

$$G_{\rm dB} = 10\log_{10} G_{\rm lin} \qquad G_{\rm lin} = 10^{G_{\rm dB}/10}$$

dB	linear	dB	linear
0	1	10	10
3	2	13	20
6	4	20	100
7	5	30	1000
−3	0.5	−10	0.1

For E-field or amplitude use 20 log not 10 log (power is squared).

Memorize: 3 dB $\Leftrightarrow \times2$, 10 dB $\Leftrightarrow \times10$. Add dB = multiply linear.

ANTENNA INTEGRATION TIPS

Ω_p integrals usually have no closed form.

Wolfram: `2*pi*Integrate[Exp[-20 t^2]*Sin[t],{t,0,pi}]`.

Narrow-beam shortcut *Gaussian beam, $\theta_{1/2} \ll 1$*

$$\Omega_p \approx \frac{\pi\,\theta_{1/2}^2}{\ln 2} \quad \text{(approximate)}$$

Only valid for narrow beams; HW4 P2 is borderline so use the integral.

Sanity check D *compare to known antennas*

$D < 2$: short / half-wave dipole. $D \sim 10$: small horn. $D > 100$: dish.

RADAR EQUATION (BONUS)

Monostatic radar *tx = rx, target RCS σ_t (m^2)*

$$P_r = \frac{P_t G^2\lambda^2\sigma_t}{(4\pi)^3 R^4} \ (\mathrm{W})$$

Doppler shift *closing velocity v_r*

$$f_d = \frac{2v_r}{\lambda} = \frac{2v_r f}{c} \ (\mathrm{Hz})$$

Factor 2 from round-trip path; v_r radial toward radar. $f_d > 0$: target approaching. $1/R^4$ falloff vs. Friis $1/R^2$.

DIPOLE

- l — antenna length (m)
- $\lambda = c/f$ — wavelength (m)
- I_0 — peak current (A)
- $\eta_0 = 120\pi \approx 377\,\Omega$
- $k = 2\pi/\lambda$ (rad/m)
- $R_{\rm rad}$ — rad. res. (Ω)
- R — obs. dist (m)

RADIATION

- $S(R,\theta)$ — pwr dens. (W/m^2)
- $S_{\max}$ — peak power density
- $F(\theta) = S/S_{\max}$ normalized
- Ω_p — solid angle (sr)
- β — HPBW (rad or deg)
- D — directivity (unitless)
- $P_{\rm rad}$ — total power (W)

ANTENNA

- $A_e = \lambda^2 D/(4\pi)$ eff. area
- A_p — phys. cross sec. (m^2)
- ξ — rad. eff. ($0 < \xi \le 1$)
- $G = \xi D$ — gain (linear)
- $G_{\rm dB} = 10\log G$
- Half-wave: $D = 1.64$, $R_{\rm rad} = 73$
- Hertzian: $D = 1.5$

FRIIS / LINK

- P_t — tx power (W)
- P_r — rx power (W)
- G_t, G_r — linear gains
- R — range (m)
- $L_{\rm fs} = (4\pi R/\lambda)^2$ path loss
- 3 dB $\to$ 2; 10 dB $\to$ 10
- $G_{\rm lin} = 10^{G_{\rm dB}/10}$

RADAR

- σ_t — RCS (m^2)
- f_d — Doppler shift (Hz)
- v_r — radial vel. (m/s)
- $P_r \propto 1/R^4$ (monostatic)
- $f_d = 2v_r/\lambda$ (round-trip)
- $c = 3\times10^8$ m/s
- $f_d > 0$: target approaching